CamJam EduKit Sensors Worksheet Three

Project Temperature Sensor

Description In this project, you will learn how to use a 1-wire temperature sensor. GPIO Zero does not yet include the 1-wire interface, so you will be using an alternative library.

Equipment Required

- Your Raspberry Pi
- 400 point breadboard
- Temperature sensor (DS18B20)
- A terminal block
- 1 x 4.7k Ω resistor
- 3 x M/F jumper wires
- 3 x M/M jumper wires

The Parts

In this circuit, you will be connecting a temperature sensor to the GPIO header of your Raspberry Pi and using Python to measure the temperature where you put the sensor. The sensor supplied in this kit is on the end of a long wire and is waterproof, which will allow you to easily measure the temperature of the room, outside a window, or even a cup of water. **Remember that electronics and water do not mix well, so keep the water away from the rest of the kit and the Raspberry Pi!**

Before you build the circuit, look at the additional parts you are going to use.

The Temperature Sensor



The sensor supplied in the kit is a 'Dallas DS18B20' sealed into a metal tube and extended with wires, and looks like the photo on the left. The sensor inside the tube looks like the device on the right.



The sensor has a '1-Wire serial' interface, which means that it sends digital messages through its output pin to the Raspberry Pi. The Pi reads these messages and puts them in a 'device file', which is like a text file. You can read this file just as you would any other text file, although you cannot edit it.

When the Raspberry Pi has received a good message from the sensor, two lines will appear in the 'device file'. The first one will end in 'YES', and the second one will end in a '*t=xxxxx*', where '*xxxxx*' is the temperature in 1/1000th of a degree Celsius. For example:

```
a3 01 4b 46 7f ff 0e 10 d8 : crc=d8 YES
a3 01 4b 46 7f ff 0e 10 d8 t=32768
```

This means that the temperature is 32.768°C.

The code in this worksheet uses an additional python library, called W1ThermSensor. Instructions for installing the library are in Worksheet 1.

The sensor has three wires (or legs). Confusingly, the colours may vary by manufacturer.

- The red wire is always the power supply (3.3v).
- If there is a black wire, then this is ground (GND), and the other wire is for the sensor data output.
- If the other two are yellow and white, then the yellow is the output from the sensor, and white is ground (GND).



The EduKit includes a terminal block because it makes connecting the temperature sensor to the breadboard a lot easier. Screw each wire of the temperature sensor into the terminal block in the order (from the left) power, data, ground.

The Resistor

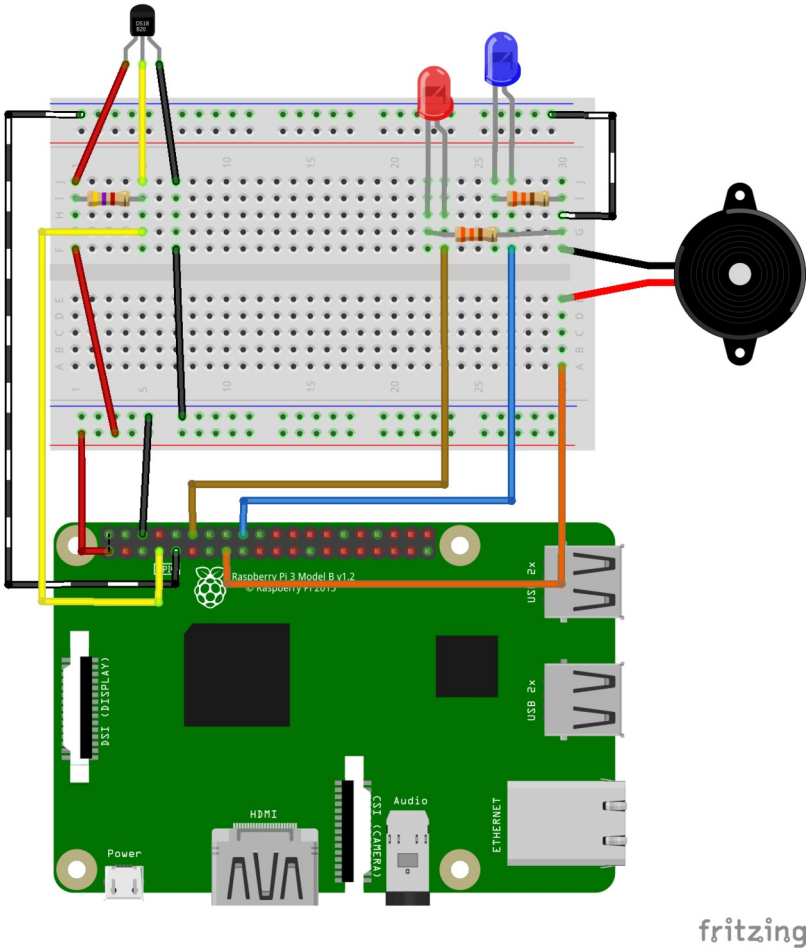


The additional resistor used in this circuit is the 4.7kΩ (or 4700Ω) resistor. You can identify the 4.7kΩ resistor by the colour bands along the body. There will be either four or five colour bands on the resistor:

- If there are four bands, the colours will be Yellow, Purple, Red, and then Gold.
- If there are five bands, the colours will be Yellow, Purple, Black, Brown, Brown.

The resistor is used as a 'pull-up' for the data line, and is required to keep the data transfer stable by supplying power to the signal circuit.

Building the Circuit



Before building this circuit, you must turn the Raspberry Pi off.

You should leave the LED and buzzer circuit connected on the breadboard, and add this new circuit to the other end.

The circuit will be using another 'ground' (GND) pin to act like the 'negative' or 0 volt ends of a battery. One of the pins marked 3v3 will provide the power for the sensor. 3v3 means that it is a 3.3-volt power supply.

Use two female-to-male jumper wires to connect the GND and 3v3 GPIO pins to the bottom two rows of holes on the breadboard. Match up the colours marked on the breadboard - red and blue - with the jumper wires from the Pi – connect 3v3 to the red row, and GND to the blue row. These two 'rails' (as they are known) will provide the ground and power supply for the whole of the breadboard.

Connect the temperature sensor as shown, with a male/male jumper wire going to the bottom 'rail' attached to the Pi's ground (GND). Connect the red wire using a jumper to the 3v3 'rail' at the bottom. This supplies the temperature sensor with its power.

The temperature sensor data lead goes into a column with one end of the 4.7k Ω resistor and another jumper wire (shown in yellow) that goes to GPIO pin 4. The program will read the temperature from this pin.

The other end of the resistor should be inserted into another column of the breadboard, between the red lead of the temperature sensor and the jumper wire connected to the 3v3 'rail'.

Code

Now it is time to write the code. Open the Python editor and type in the following code:

```
# CamJam EduKit 2 - Sensors
# Worksheet 3 - Temperature

# Import Libraries
import time
from w1thermsensor import W1ThermSensor, Unit

sensor = W1ThermSensor()

# Print out the temperature until the program is stopped.
while True:
    temp_c = sensor.get_temperature(Unit.DEGREES_C)
    temp_f = sensor.get_temperature(Unit.DEGREES_F)

    print(temp_c, " Celsius")
    print(temp_f, " Fahrenheit")

    time.sleep(1)
```

Save the file as `3-temperature.py`.

Running the Code

You are now ready to run the code. Select the Run Module menu option, under the Run menu item. Alternatively, you can just press the F5 key.

The current temperature, in degrees Celsius and Fahrenheit, should be printed on the screen every one second.

Challenge One

Measure the temperature of a glass of cold water by putting the silver end of the temperature probe into a glass of water.

Try this again with a glass of hot water, and watch the temperature change as it cools down (maybe by adding in cold water).

Challenge Two

Alter the code to light the LEDs and sound the buzzer under the following conditions:

1. Light the blue LED when the temperature is near 0°C - the sensor is accurate to 0.5°C, so light the blue LED when the temperature is at or below 0.5°C.
2. Light the red LED when the temperature is above 50°C.
3. Sound the buzzer when the temperature is above 75°C.